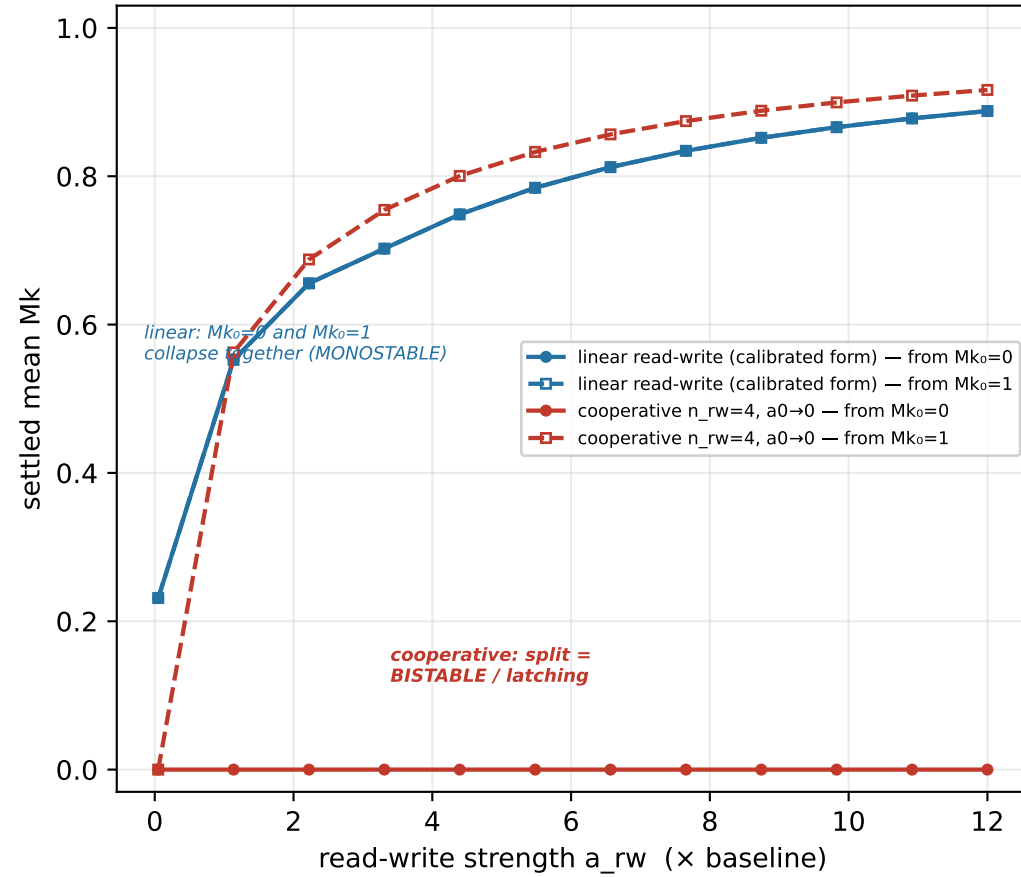


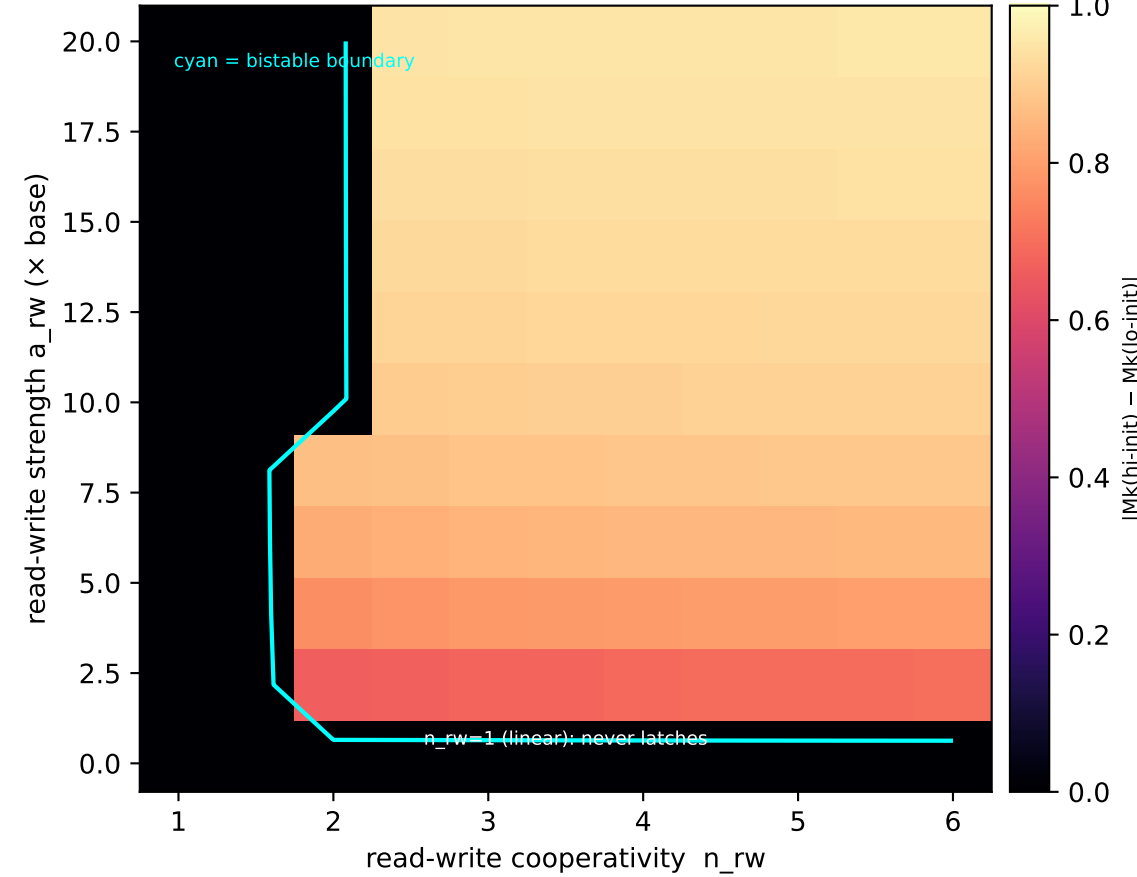
Possibility space of the *Ccnd1* H3K27me3 mark — extended bounds + structural knobs (v44.1 dynamics; not data-constrained)

Calibrated architecture is monostable (buffered). Cooperative read-write ($n_{rw} \geq 2$) + suppressed nucleation ($a_0 \rightarrow 0$) unlocks a BISTABLE latching switch coupled to cell-cycle arrest — a reachable behavior the data-fit model does not use.

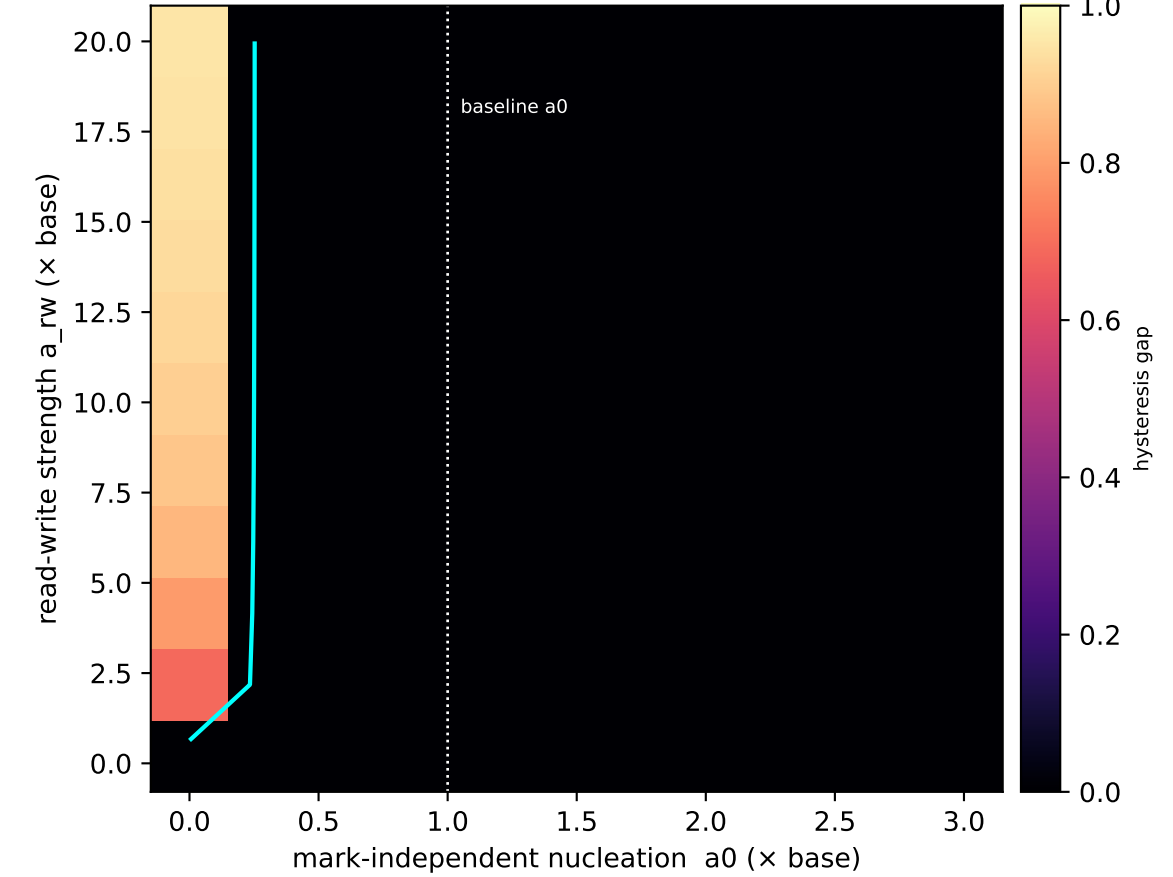
A. Extended read-write sweep: two initial conditions



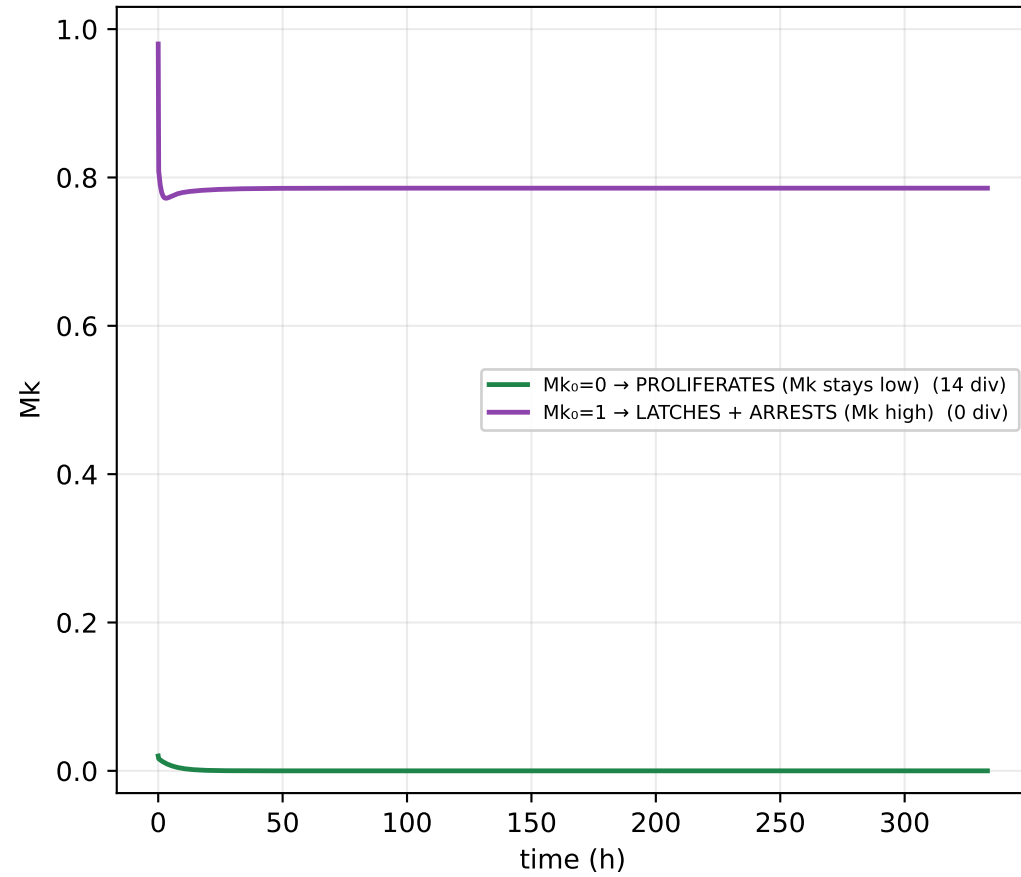
B. Latching (hysteresis gap) vs cooperativity $\times$ strength ($a_0=0$)



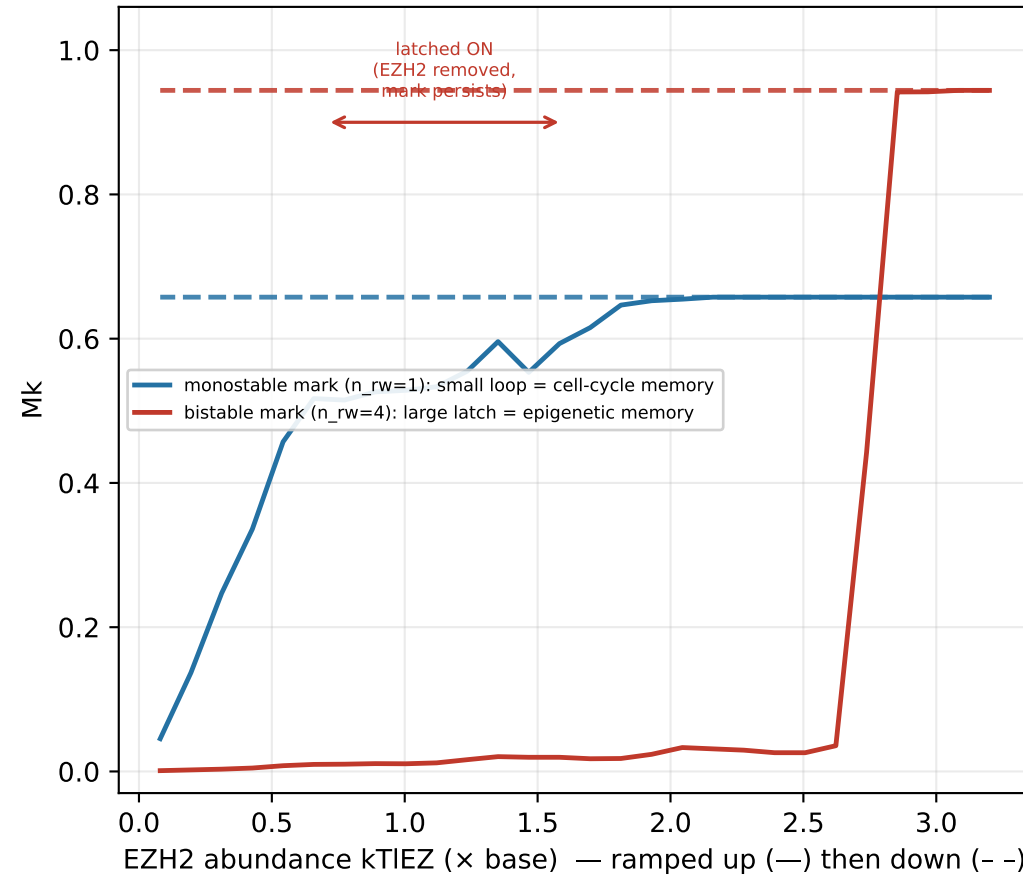
C. Latching needs low nucleation ($n_{rw}=4$)



D. Same parameters, two fates (bistable point)



E. Driver-ramp hysteresis (memory)



F. Regime map ($a_0=0$)

